Finite geometry and physical realization of complete maximal-information profiles

Astra P1 research construction
Exact rational witnesses, affine coordinates, and a physical-realization compiler

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Status: original construction in this run; unreviewed.

Scope. Finite-dimensional, explicitly supplied rational quantum channels; arbitrary finite untouched references; faithful input states; maximal operator-convex f-divergences. All polynomial bounds refer to dense matrix dimensions and rational bit lengths. They do not refer to the number of qubits, device uses, or experimental precision.

Abstract

A complete maximal-information profile records a channel's maximal f-divergence on every faithful input pair with every finite reference. Prior Astra result N1698 identifies equality of such profiles with equality of ordered normalized moments and supplies a rational greatest-channel compiler. This manuscript supplies three further constructions at that stated scope.

First, an unequal pair of rational channels admits an exactly compiled, well-conditioned separable state pair that separates the channels in one common orientation for every nonaffine operator-convex f, with gap at least $f''(1)\mu/2$ for an explicit positive rational μ. Second, a finite family has a computable affine profile dimension $p-1$ and a panel of exactly $p-1$ scalar probes that is simultaneously sufficient for every such f; the same rational affine relations apply to all reference sizes and input pairs. Third, a signed affine combination of complete profiles is physically realizable precisely when its moment functional is positive and its null space is a two-sided ideal. Equivalently, its density in the generated finite C^*-algebra has a central support projection. The criterion produces either a rational CPTP channel or a finite negative-norm or null-ideal certificate in polynomial dense-input bit complexity. An explicit qubit family exhibits a positive, nonphysical boundary profile and separates reference-free information from complete information.

These are finite mathematical constructions. They do not determine ordinary quantum capacity, supply a free divergence or optimization oracle, prove practical scalability, or achieve the programme's civilisation-level consequence anchor. The research programme remains open. Exact regression checks accompany the manuscript; they are not independent proof verification or a novelty determination.

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1 Question, result, and source contract

The finite question resolved here is:

> Given finitely many explicit rational quantum channels with a common input, determine the full affine geometry of their complete maximal-information profiles, construct physical state probes for that geometry, and decide exactly which signed affine combinations can themselves be compiled into quantum channels of some finite output dimension.

This is an exact construction question, not merely equality testing or a formal linear combination of numerical divergence values. The decisive operation is a physical-realization compiler with a finite obstruction on failure. Sections 3, 4, and 5 give the three constructions. Section 6 gives a boundary example in which positivity alone fails.

1.1 Governing prior capital

The governing Astra snapshot is commit

`c6c7966fa8ebdfb5207d2a11b6648ea95c9ecf3e`.

The current packets and native visible returns were read before construction. The complete source manifest in the accompanying packet retains exact versions, hashes where observed, reading status, and access holds.

- **N1690:** complete maximal-information evaluation and the separation between that information and ordinary quantum capacity. Its finite evaluator and paid capacity separation remain at their stated scopes. An ordinary-capacity algorithm is not supplied by it [1].
- **Corrected N1698:** complete maximal-profile equality is equivalent to equality of all ordered normalized moments and to the appropriate input-labelled, state-preserving finite-algebra isomorphism. The weight is faithful and generally nontracial. N1698 also supplies a greatest representative, a dimension classification, and a rational compiler. Its polynomial equality/compiler statements did *not* supply a polynomial inequality-witness generation bound [2].
- **Other preserved results:** the universal maximal-profile extension, finite rank criterion, dimension bounds, paid heralded recovery, N675/N557, and the hard-discrepancy/exact-marginal/relative-covariance construction retain their existing scopes. None is re-proved or enlarged by this manuscript. In particular, paid exact optimization and rational input do not become a free NP-hard optimization interface.

The extension in this manuscript is not a replacement proof of N1698. Its moment representation is used as prior capital. We prove the needed signed-linear extension, the uniform physical-probe construction, and the realization criterion. Standard finite-dimensional C^*-algebra, conditional-expectation, and GNS facts are credited rather than presented as discoveries. Likewise, short-word closure is the familiar finite linear-representation mechanism behind weighted-automaton equivalence [5]; novelty cannot be inferred merely from using it for channels.

1.2 Programme and resource status

The wider mission is civilisation-changing understanding or capability, with P=NP as the consequence anchor. The finite result here does not meet that anchor. In particular, equality of complete maximal profiles is too coarse to determine ordinary capacity, as N1690 already establishes. No efficiency consequence for arbitrary combinatorial optimization follows from polynomial algorithms on dense rational channel matrices.

P1 performs this construction. P2's autonomous quantum-processor construction and P3's independent work remain live. No reallocation, peer message, worker, descendant, experiment, purchase, publication, or source transfer is part of this packet. The coordinator retains routing/resource authority; possible future comparisons may occur only through the existing dispatcher at completed natural boundaries. No peer's unconstructed operation is used as an oracle. Sol remains held; Dots remains unmessaged. Held original PDFs and ZIPs were not retried, mirrored, or reconstructed.

2 Definitions and rational coordinates

Let $\Phi_j : \mathrm{M}_a \to \mathrm{M}_{b_j}$, $1 \le j \le k$, be completely positive trace-preserving (CPTP) linear maps. Their input-major Choi matrices are explicit over $\mathbb{Q}(i)$. Set

$$\Omega_j = \Phi_j(I_a), \qquad e_j = \operatorname{rank} \Omega_j, \qquad T_j(X) = \Omega_j^{-1/2} \Phi_j(X) \Omega_j^{-1/2}, \qquad \tau_j(Z) = \frac{1}{a} \operatorname{Tr}(\Omega_j Z).$$

All inverses involving Ω_j are taken on its support. Thus T_j is unital completely positive into M_{e_j}, and τ_j is a faithful state, usually *not* a trace. Define

$$T(X) = \bigoplus_j T_j(X), \quad \mathcal{A} = C^*(T(\mathrm{M}_a)) \subseteq \bigoplus_j \mathrm{M}_{e_j}, \quad D = \dim_{\mathbb{C}} \mathcal{A}, \quad N = \sum_j e_j^2.$$

Then $D \le N$. The coordinate maps from $\mathcal{A}$ to the jth summand are denoted π_j.

Choose a rational Hermitian traceless basis $H_1, \dots, H_q$ of the input, $q = a^2 - 1$, with $\|H_h\| \le 1$: diagonal differences and the real and imaginary off-diagonal matrix units suffice. The unit together with these letters generates the input matrix algebra as a vector space. For a word $w = h_1 \cdots h_n$, write

$$W_{w,j} = T_j(H_{h_1}) \cdots T_j(H_{h_n}), \qquad m_j(w) = \tau_j(W_{w,j}).$$

The unit moment is 1 and the one-letter moments are $\operatorname{Tr}(H_h)/a = 0$ for every channel. All nontrivial differences therefore begin at length at least two.

2.1 Complete profiles and effective supports

For faithful positive matrices S, P, define the maximal f-divergence

$$\widehat{D}_f(S\|P) = \operatorname{Tr}\left[P^{1/2} f(P^{-1/2} S P^{-1/2}) P^{1/2}\right].$$

Here $f : (0,\infty) \to \mathbb{R}$ is operator convex. The complete profile $\Pi_f(\Phi)$ is the function of *all* finite r and faithful density pairs $\sigma, \rho \in \mathrm{M}_a \otimes \mathrm{M}_r$ given by

$$\Pi_f(\Phi)(\sigma,\rho) = \widehat{D}_f((\Phi \otimes \mathrm{id}_r)(\sigma) \| (\Phi \otimes \mathrm{id}_r)(\rho)).$$

If $\rho \ge \epsilon I$, then its output is at least $\epsilon \Omega \otimes I_r$, so every such output is faithful on the fixed effective support. No finite value of $f(0+)$ is required.

Write the noncommutative perspective as

$$\mathcal{P}_f(S,P) = P^{1/2} f(P^{-1/2} S P^{-1/2}) P^{1/2}.$$

Perspective congruence gives the useful identity

$$\Pi_f(\Phi_j)(\sigma,\rho) = a(\tau_j \otimes \operatorname{Tr}_r) \mathcal{P}_f((T_j \otimes \mathrm{id}_r)(\sigma), (T_j \otimes \mathrm{id}_r)(\rho)). \tag{1}$$

The expression inside the functional belongs to $\mathcal{A} \otimes \mathrm{M}_r$ when evaluated jointly. The second tensor factor carries the ordinary, unnormalized matrix trace.

Remark 2.1 (A reference-system correction that matters)**.** For a replacer $\Phi(X) = \mathrm{Tr}(X)\omega$, the complete profile is $\widehat{D}_f(\sigma_R\|\rho_R)$, not the constant $f(1)$ in general. It equals $f(1)$ for a common reference marginal, or without a reference. When $a = 1$, all channels have this same reference-only complete profile.

These are maximal/RLD profiles. They are not measured/SLD divergences, Umegaki relative entropy, capacities, or a channel divergence defined by optimizing over different inputs for different channels.

2.2 Square-root-free word closure

For each word use raw coordinates

$$B_{w,j} = \Omega_j^{1/2} W_{w,j} \Omega_j^{1/2}.$$

The unit is Ω_j, a letter is $\Phi_j(H_h)$, and multiplication is

$$B_{uv,j} = B_{u,j}\Omega_j^{+} B_{v,j}, \qquad m_j(w) = \frac{1}{a}\,\mathrm{Tr}\, B_{w,j}. \tag{2}$$

Here Ω_j^{+} is rational. For example, if P_j is the rational orthogonal support projector,

$$\Omega_j^{+} = (\Omega_j + I - P_j)^{-1} P_j.$$

A rational column basis C for $\mathrm{ran}\,\Omega_j$ gives $P_j = C(C^*C)^{-1}C^*$.

Starting with the unit, retain an independent word whenever left multiplication by a generator leaves the current span. At most D words are retained and at most qD transitions need be considered. The retained words have length at most $D-1$. Indeed, if the span of words of length at most n equals that at length at most $n+1$, it is already invariant under every generator. This is a closure computation, not enumeration of all q^D words. Rational independence tests and ordered traces give exact moment relations.

N1698 supplies the equality implication: unequal complete profiles yield a retained word with unequal moments. The elementary argument above explains the short-word and rational computational interface used in the additional constructions.

3 A universal separable inequality witness

Theorem 3.1 (One physical witness for every nonaffine maximal f)**.** *Given two explicit rational CPTP maps with a common input, polynomial dense-input bit computation either decides their complete maximal profiles equal, or returns rational faithful separable states σ, ρ on $\mathbb{C}^a \otimes \mathbb{C}^r$, an orientation (E, F) of the two maps, and a rational $\mu > 0$, such that*

$$\begin{aligned}
&r \le e_E^2 + e_F^2,\\
&\frac{I}{2ar} \preceq \rho, \sigma \preceq \frac{3I}{2ar},\\
&\mathrm{Tr}_R\,\rho = \mathrm{Tr}_R\,\sigma = I_a/a,\\
&\Pi_f(E)(\sigma,\rho) - \Pi_f(F)(\sigma,\rho) \ge \frac{f''(1)}{2}\,\mu
\end{aligned}$$

for every nonaffine operator-convex $f : (0,\infty) \to \mathbb{R}$. The states have an explicit rational product-state mixture with $O(D)$ terms. Their reference marginals are also common. Their bit lengths, and $\log(1/\mu)$, are polynomially bounded in the dense input length.

The equality branch uses the governing finite moment criterion. The standalone witness checker verifies the unequal branch. The supplied pairwise equality output is only status metadata; it is not advertised as a separately checkable equality certificate. The finite-family checker below reconstructs word closure for that purpose.

3.1 Local RLD information controls every maximal f

For $S = P + \Delta$ of the same trace as P, put

$$\chi(S, P) = \mathrm{Tr}(\Delta P^{-1}\Delta).$$

An operator-convex f has the representation [3]

$$f(x) = f(1) + f'(1)(x-1) + c(x-1)^2 + \int_{[0,\infty)} \frac{(x-1)^2}{x+s}\, d\lambda(s), \tag{3}$$

where $c \geq 0$, λ is positive, and

$$a_f := \frac{f''(1)}{2} = c + \int \frac{d\lambda(s)}{1+s} < \infty.$$

Nonaffinity is equivalent to $a_f > 0$. This representation and its use on a compact positive spectral interval accommodate $f(0+) = +\infty$.

Lemma 3.2. *If $-\eta P \preceq \Delta \preceq \eta P$, $0 \leq \eta < 1$, and $\mathrm{Tr}\, P = \mathrm{Tr}\, S = 1$, then*

$$\frac{a_f}{1+\eta}\chi(S,P) \leq \widehat{D}_f(S\|P) - f(1) \leq \frac{a_f}{1-\eta}\chi(S,P). \tag{4}$$

Proof. The nonlinear perspective associated with (3) is

$$c\Delta P^{-1}\Delta + \int \Delta(S+sP)^{-1}\Delta\, d\lambda(s).$$

The affine term has trace $f(1)$. Since $(1+s-\eta)P \preceq S + sP \preceq (1+s+\eta)P$, inverse order bounds the integrand between the corresponding scalar multiples of $\Delta P^{-1}\Delta$. Also $1+s\pm\eta$ lie between $(1\pm\eta)(1+s)$ in the needed directions. Taking traces proves the claim; the quadratic term obeys the same looser bounds. □

Complete positivity transports the relative-order bound from an input pair to each output pair. Consequently an exactly evaluated quadratic contrast can be converted into a witness for all f with constants that do not require calling an f-evaluator.

3.2 A short word becomes a physical path probe

Let $w = AH_1 \cdots H_m B$ be a distinguishing word, with endpoints A, B in the Hermitian traceless basis. For an original word of length $n \geq 3$, take $m = n-2$. For a word AB of length two, insert the middle letter I_a and take $m = 1$. In both cases set $r = m+2 = \max\{3, n\}$. Use reference vertices $0, \ldots, m$ and one spare vertex s.

For a map T_j, define L_j and $Y_{j,\pm}$ by applying T_j to the input factor of

$$L = \sum_{h=1}^{m} H_h \otimes (|h-1\rangle\langle h| + |h\rangle\langle h-1|), \tag{5}$$

$$Y_\pm = A \otimes (|s\rangle\langle 0| + |0\rangle\langle s|) \ \pm B \otimes (\zeta|s\rangle\langle m| + \bar{\zeta}|m\rangle\langle s|), \tag{6}$$

where $\zeta \in \{1, i\}$. Unital positivity gives $\|T_j(H_h)\| \le 1$, whence $\|L_j\| \le 2$ and $\|Y_{j,\pm}\| \le \sqrt{2}$. The same bounds hold before applying a channel.

Set

$$\rho = \frac{I - tL}{ar}, \qquad \sigma_\pm = \rho + \frac{uY_\pm}{ar}, \qquad g_{j,\pm} = \frac{1}{r}(\tau_j \otimes \mathrm{Tr}_r)Y_{j,\pm}(I - tL_j)^{-1}Y_{j,\pm}. \tag{7}$$

Then $\chi(\Phi_{j,R}\sigma_\pm, \Phi_{j,R}\rho) = u^2 g_{j,\pm}$. All g values are rational and can be computed from raw output matrices without normalizing square roots.

Let $R_j = (I - tL_j)^{-1}$. Reference-block multiplication gives

$$g_{j,+} - g_{j,-} = \frac{4}{r}\Re\left[\bar{\zeta}\,\tau_j(T_j(A)(R_j)_{0m}T_j(B))\right]. \tag{8}$$

The two cross terms that survive the reference trace occupy the spare–spare block and are adjoints of one another. This is the reason for (8); no cyclic permutation under the nontracial τ_j is used.

The unique length-m path from 0 to m yields

$$(R_j)_{0m} = t^m T_j(H_1)\cdots T_j(H_m) + \mathcal{R}_j, \qquad \|\mathcal{R}_j\| \le \frac{(2t)^{m+1}}{1-2t}.$$

Write $\delta = m_E(w) - m_F(w)$, choose ζ so that $d = \left|\Re(\bar{\zeta}\delta)\right| = \max\{|\Re\delta|, |\Im\delta|\} > 0$, and choose

$$t = \frac{d}{2^{m+4}}. \tag{9}$$

Since $d \le 2$, $t \le 1/16$. The ratio of the two-channel remainder to the leading magnitude is at most

$$\frac{2(2t)^{m+1}}{(1-2t)dt^m} = \frac{1}{4(1-2t)} \le \frac{2}{7} < \frac{1}{2}.$$

Thus the contrast of the two sign contrasts is nonzero. At least one sign, with an orientation of E, F, satisfies

$$\Delta_g := g_E - g_F \ge \frac{dt^m}{r} > 0. \tag{10}$$

The compiler computes both signs exactly and chooses one. A temporary $u = 1/8$ may be used to evaluate quadratic divergences; dividing by u^2 recovers the same g.

3.3 Uniform orientation, margins, and separability

Put $S_g = g_E + g_F \ge \Delta_g$ and define

$$u = \frac{\Delta_g}{16S_g}, \qquad \eta = 2u \le \frac{1}{8}, \qquad \mu = \frac{u^2\Delta_g}{2}. \tag{11}$$

The input pair obeys

$$\left\|\rho^{-1/2}(\sigma - \rho)\rho^{-1/2}\right\| \le \frac{\sqrt{2}u}{1-2t} \le 2u = \eta.$$

Lemma 3.2 now gives

$$\Pi_f(E) - \Pi_f(F) \ge a_f u^2\left(\Delta_g - \frac{\eta S_g}{1-\eta}\right),$$
$$\frac{\eta S_g}{1-\eta} = \frac{\Delta_g}{8 - \Delta_g/S_g} \le \frac{\Delta_g}{7}.$$

This is stronger than $a_f\mu$. The norm bounds also give the displayed state-conditioning bounds in Theorem 3.1.

For explicit separability, let $\omega = I_a/a \otimes I_r/r$. If H, K are traceless Hermitian contractions, then

$$Q_\epsilon(H,K) = \frac{1}{2}\left[\frac{I+H}{a}\otimes\frac{I+\epsilon K}{r} + \frac{I-H}{a}\otimes\frac{I-\epsilon K}{r}\right] = \omega + \frac{\epsilon H\otimes K}{ar}, \quad \epsilon\in\{\pm 1\}. \tag{12}$$

These are rational product-state mixtures. Each path edge and each of the two endpoint edges contributes such a term. The inserted identity letter, when present, uses the single product state $I_a/a \otimes (I_r + \epsilon K)/r$. The total absolute coefficient weight is

$$mt + 2u \le \frac{m}{2^{m+3}} + \frac{1}{8} \le \frac{3}{16}.$$

Use the remaining nonnegative weight on ω. This supplies a concrete mixture, rather than a separability test. Tracing out the reference gives I_a/a. Endpoints are traceless, so the two states also share their reference marginal; it is I_r/r except for the inserted identity case.

All retained word lengths, matrix sizes, inversions, and rational trace operations are polynomially bounded. A nonzero rational d of polynomial height can be exponentially small, but its encoding is still polynomial. Equations (9)–(11) retain this property, including the bound on $\log(1/\mu)$. This completes Theorem 3.1.

4 Exact affine geometry and a minimal common probe panel

Theorem 4.1 (Signed-linear moment criterion)**.** *Let real $c_1,\dots,c_k$ satisfy $\sum_j c_j = 0$. The following conditions are equivalent:*

1. *$\sum_j c_j \Pi_f(\Phi_j) = 0$ on every faithful input pair and every finite reference, for one fixed nonaffine operator-convex f;*
2. *the same identity holds for every operator-convex f;*
3. *$\sum_j c_j m_j(w) = 0$ for every ordered word;*
4. *$\sum_j c_j \tau_j \circ \pi_j$ vanishes on one retained word basis of $\mathcal{A}$.*

Proof. The last two conditions are equivalent by finite word closure. If the functional vanishes on $\mathcal{A}$, its extension by the reference trace vanishes on the perspective in (1), proving the second condition.

For the other direction, fix the f in the first condition. For any faithful ρ and trace-zero Hermitian H, the coefficient of ϵ^2 in the profile at $(\rho+\epsilon H, \rho)$ is a_f times the RLD quadratic form. This follows either by analytic functional calculus near the identity or from (3). Since $a_f > 0$, the asserted profile identity gives a signed identity for every such quadratic form.

Insert the path probes (5)–(7), subtract the two signs, and vary the sufficiently small real parameter t. The coefficient of t^m in (8) extracts the real part of any ordered moment of length at least two. Using $\zeta = i$ extracts the imaginary part. Length two uses the identity insertion. The zero- and one-letter conditions already follow from the sum constraint and trace preservation. Thus every signed ordered moment vanishes. □

This argument extends the equality mechanism to signed linear relations; pairwise equality alone would not justify that extension.

4.1 The affine rank

Make a real row matrix from the constant row $(1,\dots,1)$ and the real and imaginary moment rows of a retained word basis. Let its rank be p and kernel be $K_{\mathrm{rel}} \subset \mathbb{R}^k$. Then

$$\dim \operatorname{aff}\{\Pi_f(\Phi_1),\dots,\Pi_f(\Phi_k)\} = p-1, \qquad \dim K_{\mathrm{rel}} = k-p.$$

These dimensions and the complete space of affine identities do not depend on the chosen nonaffine f. They include reference systems and all faithful input pairs, not just a selected finite sample.

Theorem 4.2 (An optimal fixed scalar panel)**.** *For a rational family, polynomial dense-input bit computation returns $p-1$ rational separable faithful probe pairs, each with reference dimension at most N, such that for every nonaffine operator-convex f the corresponding $p-1$ scalar profile values determine every profile in the entire affine span. The panel is common to all such f. It also returns p affinely independent family members and rational coefficients expressing every family member in terms of them; these same coefficients work on every faithful query, every finite reference, and every operator-convex f.*

The number $p-1$ is minimal for a fixed scalar-evaluation panel that is injective on the entire affine span, or on the relative interior of its physical mixture simplex. This is not a lower bound for identifying one member of a finite labelled list.

4.2 A uniform construction without adaptive height feedback

If $p = 1$, return an empty probe panel, any one family member as the affine basis, and coefficient 1 for every family member. All complete profiles then coincide. No minor perturbation or maximum over probe rows is needed. In the rest of this subsection assume $p > 1$.

Retain p independent real moment rows, with the first the constant row, to form $Z_0 \in \mathbb{Q}^{p\times k}$. Select p columns giving a nonsingular minor V_0. Put

$$K_0 = \left\|V_0^{-1}\right\|_\infty, \qquad M = \max\left\{1,\ n-2:\ n \text{ is a selected nonconstant word length}\right\},$$

and choose

$$t = \frac{1}{16\max\{1, K_0 p\}\, 2^{M+1}}. \tag{13}$$

For every nonconstant row, use its phase (1 or i) and its path probe. The scaled sign difference $r(g_+ - g_-)/(4t^m)$ approximates that moment row. Each entry error is at most

$$\varepsilon_t = \frac{2^{M+1}t}{1-2t}.$$

Our choice makes $K_0 p\varepsilon_t < 1$. Thus the p selected columns of the matrix of scaled sign differences and the constant row remain nonsingular. This is a fixed exact perturbation estimate from the *moment* minor, whose height was already bounded.

Collect the constant row and all individual quadratic-probe rows g_+, g_-. Their rank is at least p, by the preceding sign differences. It is at most p, by Theorem 4.1, since every row annihilates K_{rel}. Select the constant row and $p-1$ independent individual rows to form $G \in \mathbb{Q}^{p\times k}$. Choose a nonsingular column minor $V = G_J$, and put

$$K = \left\|V^{-1}\right\|_\infty, \qquad R_* = \max_{\text{selected probe rows } i} \sum_{j\in J} g_{ij},$$

$$u = \min\left\{\frac{1}{16}, \frac{1}{16K\max\{1, R_*\}}\right\}, \qquad \eta = 2u.$$

For any fixed nonaffine f, normalize a probe row by replacing its entry with

$$\frac{\Pi_f(\Phi_j)(\sigma_i, \rho_i) - f(1)}{a_f u^2}.$$

Lemma 3.2 bounds the difference from g_{ij} in absolute value by $\eta g_{ij}/(1-\eta)$. Hence the perturbation E_J of V satisfies

$$\left\|V^{-1}E_J\right\|_\infty \le \frac{K\eta R_*}{1-\eta} \le \frac{1}{7} < 1. \tag{14}$$

Every normalized f-panel has rank p, and its kernel contains K_{rel}. Equality of dimensions makes that kernel exactly K_{rel}, for every f simultaneously. The states are separable by the same mixture construction as before; here $mt + 2u \le 3/16$ again.

Finally define

$$\beta = V^{-1}G \in \mathbb{Q}^{p\times k}. \tag{15}$$

The constant row makes every column of β sum to one. Its columns on J are the standard basis. The difference between the jth coordinate vector and its J-supported β column belongs to K_{rel}. Therefore

$$\Pi_f(\Phi_j) = \sum_{\ell=1}^{p} \beta_{\ell j}\Pi_f(\Phi_{J_\ell})$$

as an identity of complete profiles, not just an interpolation identity on the selected panel.

Scalar evaluation is affine on a profile affine span of dimension $p-1$. Fewer than $p-1$ scalar evaluations have a nonzero kernel direction, which also produces distinct sufficiently close physical mixtures in an affinely independent simplex. This proves the stated minimality.

Remark 4.3 (A retained failed route)**.** An earlier adaptive prototype shrank t after estimating inverse-minor heights of the resulting probe matrix. That made a polynomial bit bound circular: shrinking t itself changes those heights. It is preserved in the packet as `adaptive_panel_prototype.py`, but it is not the governing construction or proof. Equations (13) and (14) repair this by fixing a moment basis first and perturbing its known minor, then choosing u only once from the exact quadratic panel.

No f-evaluator is used to compile the panel. Evaluating the panel for a general f, or measuring its outputs on a device, remains a separate paid operation. For the quadratic choice $f(x) = (x-1)^2$, every panel entry is rational and exactly evaluable from the supplied matrices. Signed entries of β are not physical mixing probabilities. The next section resolves that distinction.

5 Which affine profiles have a physical realization?

Fix real coefficients $c_1, \dots, c_k$ with $\sum_j c_j = 1$, and define the unital linear functional

$$\ell_c(X) = \sum_{j=1}^{k} c_j \tau_j(\pi_j X), \qquad X \in \mathcal{A}. \tag{16}$$

Here π_j is the projection to the jth channel block. The proposed profile is the signed affine expression $\sum_j c_j \Pi_f(\Phi_j)$. A physical realization means a completely positive trace-preserving map on the same input algebra, with an arbitrary finite output dimension, whose *complete* profile equals that expression. Negative c_j are allowed in the specification; they are not branch probabilities.

Theorem 5.1 (Exact physical realization criterion)**.** *The following conditions are equivalent.*

1. *Some finite-output channel Γ realizes $\sum_j c_j \Pi_f(\Phi_j)$, for one fixed nonaffine operator-convex f, on every faithful input pair and every finite reference.*
2. *A single such channel realizes the expression for every operator-convex f.*
3. *The functional ℓ_c is positive, and its zero-norm subspace*

$$\mathcal{N}_c = \{X \in \mathcal{A} : \ell_c(X^*X) = 0\} \tag{17}$$

   is a two-sided ideal of $\mathcal{A}$.
4. *Write $\ell_c(X) = \mathrm{Tr}(Z_c X)/a$, with $Z_c \in \mathcal{A}$ the density with respect to the ordinary matrix trace in the given faithful representation. Then $Z_c \ge 0$ and its support projection p_c is central in $\mathcal{A}$.*

If the original channels and c have rational real and imaginary parts, there is a deterministic polynomial algorithm in their total dense binary input length. It returns either a rational CPTP realization, or one of the following exact certificates excluding every *finite-output realization:*

$$\ell_c(P^*P) < 0, \quad \text{or} \quad \ell_c(P^*P) = 0 < \ell_c((PT(H))^*(PT(H))).$$

Here P is an explicitly given rational complex linear combination of retained words, and H is one of the fixed Hermitian input generators. The algorithm does not evaluate f.

The two conditions in item 3 have different jobs. Positivity makes ℓ_c a state. The ideal condition makes that state faithful after passing to a quotient that retains a well-defined labelled algebra of channel observables. An ordinary GNS representation exists for every positive functional; that fact alone does not provide a physical profile realization, because its distinguished vector state need not be faithful on the represented algebra.

5.1 Why every physical realization forces a quotient

Suppose Γ satisfies item 1. Apply Theorem 4.1 to the combined family $(\Phi_1, \dots, \Phi_k, \Gamma)$ with coefficients $(c_1, \dots, c_k, -1)$. The resulting identity says that the normalized ordered moments of Γ equal ℓ_c on every word.

Start with noncommutative $*$-polynomials in the Hermitian input letters. If a polynomial P evaluates to zero in the original joint algebra $\mathcal{A}$, then

$$0 = \ell_c(P^*P) = \tau_\Gamma(P_\Gamma^* P_\Gamma).$$

The state τ_Γ is faithful on the effective output support: its density is $\Gamma(I_a)/a$, which is strictly positive on that support. Therefore $P_\Gamma = 0$. Evaluation at the normalized generators of Γ consequently descends to a unital surjective $*$-homomorphism

$$\pi : \mathcal{A} \longrightarrow \mathcal{A}_\Gamma, \qquad \pi(T(X)) = T_\Gamma(X), \qquad \ell_c = \tau_\Gamma \circ \pi.$$

Positivity follows, and faithfulness gives

$$\mathcal{N}_c = \ker \pi.$$

This kernel is a two-sided ideal. This proves necessity even when the putative output dimension is unbounded in advance and the channel entries are not rational.

5.2 The density and the central-support condition

The ordinary trace restricted to a finite-dimensional $*$-algebra is a faithful inner product, so Z_c exists and is unique. Positivity of ℓ_c is equivalent to $Z_c \geq 0$. For example, a negative spectral projection of a Hermitian $Z_c \in \mathcal{A}$ belongs to $\mathcal{A}$ and would contradict positivity. For a positive Z_c, let p_c be its support. Then

$$\ell_c(X^*X) = \frac{1}{a} \left\| X Z_c^{1/2} \right\|_{\mathrm{HS}}^2, \qquad \mathcal{N}_c = \{X : Xp_c = 0\} = \mathcal{A}(1 - p_c).$$

This is always a left ideal. It is a right ideal exactly when $(1 - p_c)\mathcal{A}p_c = 0$. Taking adjoints then gives $p_c\mathcal{A}(1 - p_c) = 0$, which is equivalent to centrality of p_c. Conversely, centrality immediately makes $\mathcal{A}(1 - p_c)$ two-sided.

In a decomposition

$$\mathcal{A} \simeq \bigoplus_\lambda (M_{n_\lambda} \otimes I_{m_\lambda}), \qquad Z_c \simeq \bigoplus_\lambda (z_\lambda \otimes I_{m_\lambda}), \tag{18}$$

the criterion has a simple form: every z_λ must be either positive definite or identically zero. A nonzero density that loses rank *inside* a simple matrix block fails the criterion. A whole block may be discarded.

5.3 The channel construction

When the criterion holds, retain the central support p_c and define

$$\Gamma_c(X) = Z_c^{1/2} T(X) Z_c^{1/2} \quad \text{on } p_c\mathcal{H}. \tag{19}$$

This is completely positive. Its trace is

$$\operatorname{Tr}\Gamma_c(X) = \operatorname{Tr}(Z_c T(X)) = a\ell_c(T(X)) = \sum_j c_j \operatorname{Tr} X = \operatorname{Tr} X.$$

Its output on the identity is Z_c, and its normalized map is $X \mapsto p_c T(X) p_c$. Since p_c is central, this is the labelled quotient of T by $\mathcal{N}_c$. Its normalized state is $Y \mapsto \operatorname{Tr}(Z_c Y)/a$, which is faithful on that quotient. Its ordered moments are exactly ℓ_c, so Theorem 4.1 proves all the profile identities in item 2.

This proves the abstract equivalence. The square roots in (19) need not be rational. The following compilation gives a rational channel without assuming an exact radical or a polynomial-time integer factorization operation.

5.4 Finite certificates for positivity and the ideal condition

Let $W_1, \dots, W_D$ be a retained complex word basis of $\mathcal{A}$. Form the Hermitian Gram matrix

$$F_{ij} = \ell_c(W_i^* W_j). \tag{20}$$

Then $F \geq 0$ if and only if ℓ_c is positive on all of $\mathcal{A}$. A negative rational quadratic-form witness for F gives the first obstruction in Theorem 5.1. Exact Hermitian elimination supplies such a witness, including the case of a zero diagonal with a nonzero off-diagonal entry.

If $F \geq 0$, a rational basis of $\ker F$ gives a basis of $\mathcal{N}_c$. Test each basis element after right multiplication by each of the $q = a^2 - 1$ Hermitian input generators. A positive resulting squared norm gives the second obstruction. If every test is zero, right invariance under all generators, their products, and complex linear combinations makes $\mathcal{N}_c$ a right ideal. It is already a left ideal by positivity, so the criterion holds.

All basis words have length at most $D-1$. A negative-square certificate can therefore be written using words of length at most $D-1$, and its square has degree at most $2D-2$. A null-ideal obstruction uses a right product of degree at most D, and a squared expression of degree at most $2D$. At most qD right-multiplication tests are required. These are finite algebraic certificates, not a search through possible target output dimensions.

5.5 Compilation in raw rational coordinates

Use the direct-sum notation

$$\overline{\Omega} = \bigoplus_j \Omega_j, \qquad \overline{\Phi}(X) = \bigoplus_j \Phi_j(X), \qquad C_c = \bigoplus_j c_j I.$$

The map $\overline{\Phi}$ is CP and has trace $k \operatorname{Tr} X$; it is not a channel when $k > 1$. The corresponding channel is the uniform flagged source $\Theta = \overline{\Phi}/k$.

On effective supports, write the raw retained basis as

$$B_i = \overline{\Omega}^{1/2} W_i \overline{\Omega}^{1/2}.$$

The raw closure constructs these B_i directly, using only rational products and pseudoinverses. No square root is used to compute them. The two Gram matrices needed by the compiler and the projection target are

$$F_{uv} = \frac{1}{a}\sum_{j=1}^{k} c_j \operatorname{Tr}(B_{u,j}^{*}\Omega_j^{+}B_{v,j}), \tag{21}$$

$$H_{uv} = \operatorname{Tr}(\overline{\Omega}^{+}B_u^{*}\overline{\Omega}^{+}B_v), \qquad v_u = \operatorname{Tr}(B_u^{*}C_c). \tag{22}$$

The matrix H is nonsingular because the W_i are a basis. Solve $H\alpha = v$ and set

$$B_c = \sum_i \alpha_i B_i, \qquad K_c = \overline{\Omega}^{+}B_c\overline{\Omega}^{+}. \tag{23}$$

Indeed, if $\mathbb{E}_{\mathcal{A}}$ is the ordinary-trace conditional expectation onto $\mathcal{A}$, then

$$Z_c = \mathbb{E}_{\mathcal{A}}(\overline{\Omega}C_c), \qquad B_c = \overline{\Omega}^{1/2}Z_c\overline{\Omega}^{1/2}, \qquad K_c = \overline{\Omega}^{-1/2}Z_c\overline{\Omega}^{-1/2}$$

on support. The normal equations (22) are exactly the orthogonal-projection equations for this Z_c. Thus K_c is rational and positive semidefinite whenever the positivity test succeeds.

Factor $K_c = C^{*}C$ with a rectangular rational complex C, as explained below, and emit

$$\widetilde{\Gamma}_c(X) = C\overline{\Phi}(X)C^{*}. \tag{24}$$

It is rational, CP, and TP. To verify its normalized structure, put $M = C\overline{\Omega}^{1/2}$. Then $M^{*}M = Z_c$. Its polar decomposition is $M = UZ_c^{1/2}$ with $U^{*}U = p_c$. Equation (24) is an isometric embedding of (19); therefore it has the required complete profile. Pseudoinverses and the factorization can operate in the original output spaces, retaining zero-support coordinates harmlessly.

Rational Gram factorization. Exact Hermitian LDL^{*} elimination reduces the factorization to positive rational pivots $d = n/m$. Write the integer nm in binary. An even power of two is one square; an odd power of two is the sum of two equal squares. After division by m^2, this gives

$$d = \sum_{\nu} r_\nu^2, \qquad r_\nu \in \mathbb{Q},$$

with at most twice the bit length of nm summands and polynomial bit length. Expanding each pivot row accordingly gives $C^{*}C = K_c$. This elementary rectangular construction is prior N1698 compiler capital; it is used here for a different signed-functional realization problem. It does not promise the smallest rational output dimension.

5.6 A paid heralded implementation from the supplied family

The rational channel construction also specifies an exact success operation on the uniform flagged source $\Theta(X) = \bigoplus_j \Phi_j(X)/k$. Define

$$p = \frac{1}{k\operatorname{Tr}K_c}, \qquad \mathcal{S}(Y) = pk\,CYC^{*}. \tag{25}$$

Because $K_c \geq 0$ and $K_c \leq (\operatorname{Tr}K_c)I$, this is trace-nonincreasing. Its failure effect is the rational positive matrix $I - pkK_c$, and

$$\mathcal{S}\circ\Theta = p\widetilde{\Gamma}_c. \tag{26}$$

The identity holds as a CP-map identity, including on entangled reference systems. The success probability is the same p for every input. Trace preservation of $\widetilde{\Gamma}_c$ implies $k\operatorname{Tr}K_c \geq 1$, so $0 < p \leq 1$.

For a fully rational instrument, express the positive scalar pk as a sum of rational squares $\sum_\nu s_\nu^2$ and use success Kraus operators $s_\nu C$. These give *the same* output channel in (26). Factoring an arbitrary replacement Gram matrix without this output check could produce a different isometric encoding. Factor the failure effect to obtain a failure map to a fixed orthogonal flag.

For this particular filter, the largest permitted success scaling is

$$p_{\mathrm{filter}} = \frac{1}{k\,\|K_c\|_{\mathrm{op}}}.$$

The rational p in (25) is a certified lower choice. No optimality over all channel-conversion protocols is asserted. For n adaptive target calls, the all-success branch has probability p^n and is exactly the target branch. Full-block attempts with known resettable inputs have expected cost n/p^n source calls if every attempt consumes n calls. Failure can destroy an unknown input; this calculation does not authorize retrying unknown quantum data. The factor k is essential because the available flagged *channel* is $\overline{\Phi}/k$.

5.7 Consequences for the physical region and minimum dimension

Within the finite affine functional space generated by the supplied family, the realizable region is convex. If two positive densities have central support projections, a strictly positive weighted sum has the join of those supports, again central. Its closure is precisely the positive affine slice. To see density, mix any positive ℓ_c with

$$\tau_0 = \frac{1}{k}\sum_j \tau_j \circ \pi_j.$$

The state τ_0 is faithful on $\mathcal{A}$. Hence $(1-\epsilon)\ell_c + \epsilon\tau_0$ is faithful and realizable for every $\epsilon > 0$, and tends to ℓ_c.

The physical region is thus the union, over central projections, of the positive densities that are strictly positive on that central support and zero elsewhere, intersected with the specified affine slice. It need not be closed. These closure statements concern finite moment coordinates, equivalently a fixed complete-profile panel, or pointwise convergence at each fixed faithful query. They do not assert uniform convergence over all reference sizes and all nearly singular input states. If $\mathcal{A}$ is commutative, every support projection is central: positivity alone suffices, and the region is closed. A noncommutative algebra permits the obstruction but does not force every affine slice to exhibit it.

The quotient $\mathcal{A}/\mathcal{N}_c$ also fixes the smallest possible complex output dimension. In (18), let the active indices have $z_\lambda > 0$. Then

$$b_{\min} = \sum_{\lambda\,\mathrm{active}} n_\lambda. \tag{27}$$

Necessity follows because every realization gives a faithful representation of this quotient, and a faithful representation must contain each active simple summand. Sufficiency follows by absorbing multiplicity into $h_\lambda = m_\lambda z_\lambda$ and using

$$X \longmapsto \bigoplus_{\lambda\,\mathrm{active}} h_\lambda^{1/2} T_\lambda(X) h_\lambda^{1/2}.$$

This is the minimum faithful-representation argument and greatest-channel structure already supplied by corrected N1698, now applied to the newly determined quotient. The rational compiler above can have a larger output. Equation (27) is not an additional polynomial guarantee for a minimum-dimensional rational encoding.

Corollary 5.2 (Finite physical interpolation with an explicit promise)**.** *Suppose a target complete profile is promised to lie in the affine span of a supplied rational family, and its exact values on the quadratic panel of Theorem 4.2 are supplied as rational numbers. Then polynomial dense-input computation either produces a rational physical channel with that entire profile, or produces a finite obstruction proving that no finite-output channel has that profile.*

Proof. The constant row and the panel invert the affine coordinate system. Solve for rational coefficients c with $\sum c_j = 1$, then apply Theorem 5.1. Redundant coefficient choices give the same functional by Theorem 4.1, so the answer is independent of that choice. □

The affine-span promise and the exact rational values are part of the input contract. A finite panel cannot certify that an arbitrary unknown device lies in the promised span, and noisy measured values are not exact rational profile data.

6 An explicit rational qubit family and a nonphysical positive boundary

The distinction between a positive functional and a physically realizable complete profile occurs already in a two-dimensional simple algebra. The example also shows that reference-free tests can miss an entire affine degree of freedom.

Write $\sigma_x, \sigma_y, \sigma_z$ for the Pauli matrices and fix $\kappa = 1/4$. Define the unital, trace-preserving CP map

$$T(X) = \frac{\operatorname{Tr} X}{2} I_2 + \frac{\kappa}{2} \operatorname{Tr}(X\sigma_x)\sigma_x + \frac{\kappa}{2} \operatorname{Tr}(X\sigma_y)\sigma_y. \tag{28}$$

Thus $T(\sigma_x) = \kappa\sigma_x$, $T(\sigma_y) = \kappa\sigma_y$, and $T(\sigma_z) = 0$. Complete positivity follows, for example, from its Choi matrix: the only nontrivial 2×2 principal block has eigenvalues $1/2 \pm \kappa$. The algebra generated by its range is M_2.

For $-1 < s < 1$, put

$$D_s = \begin{pmatrix} (1+s)/2 & 0 \\ 0 & (1-s)/2 \end{pmatrix}, \qquad \Gamma_s(X) = 2D_s^{1/2} T(X) D_s^{1/2}. \tag{29}$$

This is a channel because $\operatorname{Tr}(D_s T(X)) = \operatorname{Tr} X/2$. In input-major Choi order its matrix is

$$J_{\Gamma_s} = \begin{pmatrix} d_0 & 0 & 0 & \kappa\sqrt{1-s^2} \\ 0 & d_1 & 0 & 0 \\ 0 & 0 & d_0 & 0 \\ \kappa\sqrt{1-s^2} & 0 & 0 & d_1 \end{pmatrix}, \qquad d_0 = \frac{1+s}{2}, \quad d_1 = \frac{1-s}{2}. \tag{30}$$

The relevant determinant is $(1-s^2)(1/4-\kappa^2) > 0$, and the partial trace is I_2. Its normalized map is exactly T, and its normalized state is

$$\tau_s(W) = \operatorname{Tr}(D_s W).$$

Choose the two rational channels Γ_0 and $\Gamma_{3/5}$ as the supplied family. Their off-diagonal Choi entries use the rational numbers 1 and 4/5. The affine functional

$$\ell_s = \left(1 - \frac{5s}{3}\right)\tau_0 + \frac{5s}{3}\tau_{3/5} = \operatorname{Tr}(D_s\,\cdot) \tag{31}$$

is positive exactly for $-1 \le s \le 1$, and has a central support projection in M_2 exactly for $-1 < s < 1$. Consequently the realizable portion of this affine complete-profile line is the *open* interval

$$-1 < s < 1,$$

while its positive closure is $[-1,1]$. The endpoints are positive formal profiles with no finite-output physical realization. This assertion quantifies over every candidate channel, not merely the displayed parametrization.

6.1 A signed combination that is physically attainable

At $s = 4/5$, the coefficients are $(-1/3, 4/3)$ and $\sqrt{1-s^2} = 3/5$. Therefore the rational qubit channel $\Gamma_{4/5}$ satisfies, for every nonaffine operator-convex f, every reference, and every faithful query,

$$\Pi_{\Gamma_{4/5},f} = -\frac{1}{3}\Pi_{\Gamma_0,f} + \frac{4}{3}\Pi_{\Gamma_{3/5},f}. \tag{32}$$

This is a relation between complete profiles, not an instruction to subtract physical channels or sample a negative probability. The compiler produces a physical representative and a separate success operation. On this input its automatically emitted rational representative has output dimension 14, effective output rank 4, and certified success probability $8/29$ from the uniform flagged source. The explicit channel in (29) shows that the minimum complex output dimension is 2. The larger compiler output is a cost of the chosen rational factorization, not a necessity theorem.

6.2 A positive endpoint with a finite obstruction

At $s = 1$, the coefficients are $(-2/3, 5/3)$ and the functional is the vector state $\ell_1(W) = \langle 0|W|0\rangle$. Let

$$P = |1\rangle\langle 1| = \frac{1}{2}\left(I + i\kappa^{-2}T(\sigma_x)T(\sigma_y)\right).$$

Then

$$\ell_1(P^*P) = 0, \qquad \ell_1\big((PT(\sigma_x))^*(PT(\sigma_x))\big) = \kappa^2 > 0. \tag{33}$$

The first equality forces P to vanish in any faithful physical realization. Multiplying its image by $T_\Gamma(\sigma_x)$ would still give zero, contradicting the second equality. This is the null-ideal obstruction without a spectral decomposition or an unknown-output search. The included compiler emits a differently scaled word certificate with positive product norm $1/64$.

For $s = 6/5$, positivity already fails: $\ell_{6/5}(P^*P) = -1/10$. The corresponding regression uses another negative polynomial square, with value $-11/6400$. The two obstruction types are mathematically distinct and are tested separately.

6.3 Why the limiting channel does not realize the limiting profile

As $s \uparrow 1$, the channels Γ_s converge to the pure-output replacer

$$\Gamma_1(X) = (\operatorname{Tr} X)|0\rangle\langle 0|.$$

This limiting channel has a smaller effective support. For the faithful input states

$$\rho = I_2/2, \qquad \sigma = (I_2 + h\sigma_x)/2, \qquad 0 < h < 1,$$

one computes for every $-1 < s < 1$

$$\widehat{D}_f(\Gamma_s(\sigma)\|\Gamma_s(\rho)) = \frac{1}{2}\left[f(1+h\kappa) + f(1-h\kappa)\right]. \tag{34}$$

At the endpoint channel Γ_1, the value is $f(1)$. For the quadratic generator $f(x) = (x-1)^2$ and $h = 1/2$, the values are $1/64$ and 0, respectively. Thus convergence of channels through a loss of output rank need not commute with taking these maximal-information profiles. This

concrete discontinuity is part of the example; no claim to the first discovery of singular-boundary discontinuities is made.

There is also a fully rational sequence approaching the endpoint. Set

$$s = \frac{1-t^2}{1+t^2}, \qquad \sqrt{1-s^2} = \frac{2t}{1+t^2}, \qquad t > 0 \text{ rational},$$

and let t tend to zero. Every channel in the sequence has rational Choi entries, yet the positive limit of its affine profile coordinates is not physically realizable. For any fixed faithful query the formal profile values have a finite affine limit. There is no uniform assertion over input states whose eigenvalues also approach zero.

6.4 An entire degree of freedom invisible without a reference

More is true than (34): all the channels Γ_s, $-1 < s < 1$, have the same *reference-free* maximal-f profile for every f in the stated class. To verify this, let $R = T(\rho)$ and $S = T(\sigma)$ for arbitrary faithful qubit states. Both lie in $\mathrm{span}_{\mathbb{R}}\{I, \sigma_x, \sigma_y\}$ and have trace 1. The positive 2×2 matrix $R^{-1/2}SR^{-1/2}$ has at most two eigenvalues, so there are scalars α, β, depending on R, S, f but not on s, such that

$$f(R^{-1/2}SR^{-1/2}) = \alpha R^{-1/2}SR^{-1/2} + \beta I.$$

The perspective is therefore $\alpha S + \beta R$. Since $\mathrm{Tr}(D_s R) = \mathrm{Tr}(D_s S) = 1/2$, the divergence is $\alpha + \beta$, independent of s.

Complete profiles nevertheless distinguish s, because the ordered moment

$$\tau_s(T(\sigma_x)T(\sigma_y)) = i\kappa^2 s$$

does. Theorem 3.1 turns this imaginary second moment into an explicit reference-assisted, separable, faithful input pair. A reference is essential for this distinction even though entanglement of the two input states is unnecessary. The state on the normalized algebra is not tracial, so discarding imaginary ordered moments would lose real information.

7 Interfaces and complete cost accounting

7.1 What is supplied and what is returned

The algorithms take *explicit dense rational Choi matrices*, with a common input dimension. Their dimensions, the bit lengths of their entries, and (for realization) the rational affine coefficients are part of the input size. The code uses $\mathbb{Q}(i)$ arithmetic and rejects floating-point entries, nonpositive Choi matrices, and trace-preservation failures. Input-major Choi ordering is recorded in the packet README.

The three interfaces are distinct:

1. **Inequality witness.** Given two channels, decide profile equality by prior finite closure. On inequality return the word, rational faithful separable input states, their explicit product-state mixtures, the orientation, and a positive rational μ. The guarantee is simultaneous over the entire stated f class. No call to an f evaluator occurs in the compiler.
2. **Affine geometry.** Given a finite family, return its dimension, a basis of every affine relation, independent family representatives, rational reconstruction coefficients, and a common panel of $p - 1$ physical input pairs. For the quadratic generator, every panel value is rational and directly computable. For a general f, numerical evaluation, accuracy and conditioning of the resulting coordinate inversion are additional tasks.

3. **Physical realization.** Given a finite family and rational c with $\sum_j c_j = 1$, return a rational CPTP Choi matrix, a rectangular rational factor of its filter Gram matrix, and finite moment verification data, or return a finite polynomial obstruction. A NO certificate excludes *all* finite output dimensions. A YES certificate does not promise the smallest rational output dimension, a short gate sequence, or a practically efficient instrument.

These inputs are not experimental black boxes, compact quantum circuits, implicit optimization polytopes, arbitrary functions given by names, or samples from an unknown device. Translating any of those representations into this interface requires its own cost and validity argument. A general operator-convex f need not have a supplied computable evaluator; the uniform theorem does not manufacture one.

7.2 Why the dense-input bit bounds are polynomial

Write $q = a^2 - 1$, $N = \sum_j e_j^2$, and $D = \dim_{\mathbb{C}} \mathcal{A} \le N$. Closure retains at most D independent words, of length at most $D - 1$, and tests at most qD generator extensions after the unit. Raw multiplication uses rational matrices and the support pseudoinverses. A common-denominator bound for words of length at most D gives polynomial bit length: multiplication adds entry heights, and matrix summation adds only a logarithmic factor per multiplication. Rank tests and linear solves can use fraction-free exact elimination, so determinants and their minors give polynomial output-height bounds. Thus the field-operation counts and the required integer bit lengths are both polynomial in the dense input.

The affine construction uses fixed minors of already bounded-height matrices. Its two perturbation parameters are explicit rational reciprocals; the proof never repeatedly halves a parameter while re-estimating a height that itself depends on that parameter. This is why the unsuccessful adaptive prototype is not used for the complexity claim.

For realization, forming the $D \times D$ moment and Hilbert–Schmidt Gram matrices takes polynomially many exact matrix operations. Positive-semidefiniteness, kernels, the density solve, and at most qD right-generator null tests are polynomial exact linear algebra. A single $D \times D$ solve has cubic arithmetic complexity using ordinary elimination; this is *not* a cubic bound for the full channel algorithm, which also contains matrix products, closure tests, and rational bit arithmetic.

If $R = \sum_j b_j$ and H bounds the bit heights of rational positive pivots in the filter factorization, the binary sum-of-squares construction uses $O(RH)$ output rows. It uses no integer-factorization oracle and no irrational square roots. The emitted dense Choi matrix therefore has polynomial length. A rational Kraus description can also be obtained by applying the same factorization method to that Choi matrix. None of this asserts efficient exact synthesis over a prescribed finite universal gate alphabet.

7.3 Compilation is separate from observation and control

The rational numbers t, u, μ and the heralding probability can be exponentially small in their bit-description lengths. Faithfulness and bounded condition number of the probe states do not make the output difference large. There is no polynomial sample-complexity conclusion hidden in the polynomial compiler.

For a direct operational illustration, compare the identity channel with

$$\Psi_\varepsilon = (1 - \varepsilon)\,\mathrm{id} + \varepsilon \mathcal{R}, \qquad \varepsilon = 2^{-L},$$

where $\mathcal{R}$ is a fixed rational replacer. Their half diamond distance is at most ε. Replacing one use at a time in an arbitrary adaptive n-use protocol bounds the final half trace distance by $n\varepsilon$. A constant distinguishing advantage can consequently require $\Omega(2^L)$ uses even though the channel description and the exact witness have polynomial bit length. This follows from the usual

hybrid/contractivity argument and does not assume a particular measurement or a particular divergence estimator.

Likewise, the heralded realization described in Section 5 pays for the uniform flag channel, the filter, its success record, and failure handling. For n uses of the success branch the specified construction has all-success probability p^n. Restarting a complete known-input block costs n/p^n expected channel uses. This restart statement is unavailable for an unknown quantum input that cannot be recreated after failure. Directly synthesizing the emitted channel may be preferable; its actual control cost is a separate implementation question. The probability quoted here is the certified probability for the supplied filter, not an optimum over every possible implementation.

7.4 Equally informed alternatives and non-claims

An alternative given the same dense rational input may use matrix-algebra decomposition, exact linear algebra, semidefinite methods where their hypotheses apply, or a greatest-representative compiler from N1698. The present result does not establish a runtime advantage over all such alternatives. Its specific enabling operation is the complete YES/NO realization criterion together with physical probes and finite certificates; the additional obstruction is closure of zero-norm relations under multiplication.

The algorithms are polynomial in a, b_j and entry heights. Since $a = 2^{n_A}$ for n_A input qubits, dense-input polynomial time can already be exponential in the number of qubits. No NP-hard optimization interface, polynomial ordinary-capacity evaluation, or general polynomial quantum-circuit compilation is provided. N1690's evaluator/capacity separation remains operative: a complete maximal-information profile does not in general determine ordinary quantum capacity. N675's differing input/interface and reference-cutoff obstructions are not contradicted by an algorithm on the explicit finite moment representation used here.

8 Original-source comparison, verification, and remaining programme

8.1 Comparison with the governing capital and primary literature

The literature comparison was used before committing to the construction, and then again on the mechanisms used in the proof. The following distinctions prevent a change of terminology from being mistaken for a new operation.

Source	Supplied mechanism	Role and boundary in this construction
N1698, corrected governing packet [2]	Ordered normalized moments, equality classification, greatest representative, rational compiler.	Used as prior capital. The new witness construction supplies a polynomial generation bound not supplied by the earlier equality statement. Signed-linear geometry and the affine physical-realization decision are separately proved here.
N1690 [1]	Complete maximal-information evaluator; channels with equal profiles and different ordinary capacities.	Fixes the consequence boundary. Profile geometry does not remove the ordinary-capacity problem. No prior held proof/code packet was acquired or reconstructed.

Source	Supplied mechanism	Role and boundary in this construction
Hiai–Mosonyi [3]; Matsumoto [4]	Maximal quantum f-divergences, operator perspectives, integral representations and reversibility distinctions.	The perspective and Löwner representation are credited. The proof uses integrability at 1 and does not impose an unnecessary finite-$f(0)$ condition. It makes no transfer to Umegaki or measured/SLD divergence without a separate argument.
Kiefer et al. [5]	Rational weighted-automaton equivalence, finite linear closure, short distinguishing words and minimisation.	Finite word closure is established machinery. The operation earned here is its conversion into faithful separable channel probes with a uniform all-f sign, and into exact realization certificates.
Volčič [6]	Free rational positivity, finite truncations, GNS/domain extensions, sums of squares on free spectrahedra.	Positivity and finite representations already have constructive precedents. A positive cyclic GNS state need not be faithful; that is precisely why positivity alone does not yield a channel with the requested complete profile here. The null-ideal condition is proved, not assumed.
Hirche [7]	Channel-order relations through BKM and measured/SLD information, smoothed/restricted orders, signed lifts, and capacity consequences with additional assumptions.	The original hypotheses were checked. A signed Hermiticity-preserving lift is not automatically CPTP, and measured/SLD or BKM information is not the maximal/RLD functional used here. Its capacity results do not supply a missing paid evaluator or control operation.

The comparison is bounded. It establishes which mechanisms have been used and which tempting transfers are invalid; it does not establish worldwide historical priority. A 2026 record on continuity/testing regions and a 2026 record on operator-valued maximal divergences were encountered, but their full originals were not successfully read. They are recorded as metadata-only or held in the source manifest, not treated as supporting evidence or cleared novelty comparisons. Public Zenodo records were also inspected during destination exploration. Their topic labels were not used as compatibility evidence, and none is a mathematical premise of the theorems above.

8.2 What the accompanying computation actually verifies

The packet contains three compilers, three certificate checkers, exact rational fixtures, and regression scripts. The checkers replay certificates from emitted matrices and states, including positivity, trace preservation, product-mixture decomposition, moment relations, and the relevant sign or null-ideal condition. They share the exact-arithmetic support module and some construction utilities. They are therefore useful reproducibility checks, not independent implementations or

formal proof verification.

The recorded suites contain the following cases:

- **Eight witness cases:** identity versus replacer; an imaginary second-moment distinction; singular-output embeddings; a deliberately supplied longer path; two equality cases; rejection of a non-CP input; and rejection of a non-TP input. The longer path tests the path construction and is not claimed to be a shortest closure witness.
- **Four panel cases:** a six-channel family of affine dimension three; a signed reconstruction; a family of identical complete replacer profiles; and a one-dimensional input. The latter two return no probes. Missing-probe and altered-coefficient certificates are rejected.
- **Five realization cases:** a signed interior YES; the positive nonphysical boundary NO; a negative-density NO; a central-quotient YES; and a one-dimensional-input YES. Altered certificates are rejected. The interior certificate emits a 14-dimensional rational channel with effective support dimension four, whereas the example also has a two-dimensional complex realization. This gap is disclosed, not hidden as a minimality result.

All 17 recorded cases pass their specified checks. In the positive nonphysical boundary fixture the checked null norm is 0 and the checked right-product norm is $1/64$. In the negative-density fixture the emitted negative norm is $-11/6400$. The six-channel panel uses three reference-assisted probes of reference dimension three and verifies two independent affine relations. The signed-interior realization records success probability $8/29$ for its stated filter. These are exact finite identities; the recorded fractions depend on the explicit fixture normalization.

Reported small-instance times are all below one second for the recorded individual compiler or checker runs. Such timings are neither a complexity proof nor a scalability experiment. No laboratory device was tested; no hardware, spending, worker, extra researcher, or outreach was used. The correctness arguments are the proofs above. The regression logs are checks of those arguments on the included finite cases.

8.3 Useful failures retained

Three failed shortcuts shaped the construction and remain part of the record.

1. **Adaptive panel height feedback.** Repeatedly shrinking a probe parameter while bounding a minor that depends on that parameter did not establish a polynomial height bound. The file `adaptive_panel_prototype.py` is retained as a deprecated prototype. The fixed-moment-minor construction in Section 4 replaces that argument.
2. **Positivity as physical realization.** A positive moment Gram matrix alone can yield a nonfaithful cyclic state whose zero-norm relations do not survive multiplication. The boundary example proves the failure exactly. The two-sided-ideal test repairs the operation without borrowing a peer's unconstructed realization map.
3. **Reference-free reconstruction.** The qubit example has an entire parameter invisible to every reference-free maximal-f query. The complete representation must retain ordered complex moments and reference-assisted probes. Dropping imaginary moments or substituting a tracial weight would erase the distinction.

Earlier useful failures and accepted constructions remain preserved at their original scopes. In particular, this packet does not revisit or extend N675/N557, the prior rounding result, the universal extension, or the previously paid recovery and capacity-source work.

8.4 Consequences earned and the larger unresolved objective

The earned deductions are exact: finite affine dimension and relations; a minimum fixed scalar panel over the affine span; a common all-f physical inequality witness; a finite rational physical-realization compiler or obstruction; the positive-cone/physical-subset distinction; closure by faithful

mixtures; the minimum complex output dimension at the algebraic scope; and the explicit qubit separation between reference-free and complete profiles. These deductions are mutually compatible because they use the same generated algebra, its faithful source weights, and exact rational input.

They do not by themselves deliver the programme's required civilisation-changing capability. A route to that consequence must supply an additional operation with a materially stronger interface or cost guarantee, or a broad explanatory result that genuinely reaches the consequence anchor. Ordinary quantum capacity, practical device-scale inference, and a polynomial solution of arbitrary NP problems remain unresolved here. It would be scientifically incorrect to present this finite construction as a completed substitute for that larger objective.

The immediate scientific organization remains independent work: P2's live quantum-processor question and P3's independent construction are preserved. This result supplies no necessary missing oracle for either role, so no allocation change is justified by it alone. Any later scientifically useful comparison must carry the completed original source, its scope and costs, through the coordinator's existing dispatcher. Publication, if separately authorized, does not constitute verification.

Status: original construction in this run; unreviewed. No independent review or formal verification has been obtained. Source access and transfer holds remain unchanged. The complete larger research programme remains open.

References

[1] Astra P1, *Exact maximal-information evaluator and paid zero-versus-positive capacity separation*, governing current packet N1690, pinned Astra commit `c6c7966fa8ebdfb5207d2a11b6648ea95c9ecf3e`. The exact content version is recorded without alteration in the source manifest. https://github.com/harjassand/astra-research-knowledge/blob/c6c7966fa8ebdfb5207d2a11b6648ea95c9ecf3e/knowledge/current/packets/N1690.json. Native visible return read; held report/ZIP/code remain unacquired.

[2] Astra P1, *Greatest quantum channels determined by complete maximal-information profiles*, corrected governing current packet N1698, same pinned Astra commit. Exact content version and visible-return coverage are retained in the source manifest. https://github.com/harjassand/astra-research-knowledge/blob/c6c7966fa8ebdfb5207d2a11b6648ea95c9ecf3e/knowledge/current/packets/N1698.json. Polynomial equality/compiler guarantees do not supply a polynomial witness-generation bound. Held originals remain unacquired.

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[7] Christoph Hirche, *Quantum estimation, channel orders, and private capacity*, arXiv:2610.03706, 2026; 40-page PDF inspected. https://arxiv.org/pdf/2610.03706. The source distinctions recorded in Section 8 are part of the evidence contract; no claim is made that its signed lifts are CPTP or that its measured/SLD and BKM quantities equal the maximal divergence.